

LEE YEONG, KIM

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Education

- 01/03/2016 – **Combined M.S. and Ph.D. in Physics**, grade 4.02/4.3
19/08/2022 *Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea*
Research advisor: Prof. Bum Suk Zhao (zhao@unist.ac.kr)
Dissertation title: Study on matter-wave optics under grazing incidence condition with solid surfaces
- 01/03/2011 – **Bachelor of Science in Physics**, grade 3.60/4.3
23/02/2016 *Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea*
Minor major: Energy Engineering
Full scholarship recipient

Education and Research Experience

- 01/01/2026 – **Postdoctoral Researcher** | *University of Vienna* Austria
Present Research advisor: Prof. Markus Arndt
- Studying matter-wave interference of biomolecules in the BIOTIMA project, extending interferometry from peptides (> 2,000 Da) towards intact proteins
- Supported by the quantA Postdoctoral Fellowship (2026–2027)
- 18/07/2023 – **Career transition period** in Zürich Switzerland
31/12/2025 Spouse's postdoctoral stay
- Remained research actively: two peer-reviewed publications (2024) and contributions to MOLEC 2024 and ISMB 2025
- Prepared fellowship applications leading to the quantA Postdoctoral Fellowship and an MSCA Seal of Excellence (proposal score 95.8/100)
- 01/09/2022 – **Postdoctoral Researcher** | *UNIST* Korea
30/06/2023 Research advisor: Prof. Bum Suk Zhao
- Reveals effects of macroscopic surface curvature, grating magnification, and beam emergence in grazing incidence atom scattering, guiding two graduate students (*Phys. Rev. A* **110**, 013313)
- Analyzed diffractive mirrors for neutral-atom matter-wave optics (*Fara. Dis.* **251**, 160)
- 2014–2019 **Visiting Student** | *Fritz-Haber-Institut der Max-Planck-Gesellschaft* Berlin, Germany
Research advisor: Dr. Wieland Schöllkopf and Prof. Bum Suk Zhao
Seven visits: 4 months (2014, 2017), 2 months (2015, 2016), and 10 months (2019).
- Carried out ^4He , $^4\text{He}_2$, $^4\text{He}_3$ and D_2 scattering experiments on the FHI molecular-beam apparatus
- Experimentally tested Babinet's principle in matter-wave diffraction and identified multiple-edge diffraction, a non-destructive mechanism that enhances elastic reflection from microstructured surfaces (*PCCP* **23**, 8030; *PCCP* **24**, 21593 – front cover)
- Observed the transition from quantum reflection to edge diffraction in a periodic array of half planes (*Phys. Rev. Lett.* **122**, 040401 – Editor's Suggestion)
- 02/03/2014 – **Undergraduate Research Assistant** | *UNIST* Korea
23/02/2016 Research advisor: Prof. Bum Suk Zhao
- Built Monte-Carlo trajectory simulations of molecules deflected by pulsed non-resonant optical standing waves, showing that rotational-state-dependent polarizability governs the optical dipole force (*Phys. Rev. Lett.* **115**, 223001)

Awards and Honors

- 17/02/2026 Seal of Excellence, Marie Skłodowska-Curie Actions Postdoctoral Fellowship, European Commission | Proposal score: 95.8/100
- 16/02/2023 UNIST Best Research Award, UNIST | Top 5 excellent graduates
- 11/01/2021 Best poster presentation Award, The Korean Physical Society
- 28/04/2021 Best poster presentation Award, The Korean Physical Society
- 22/04/2016 Best oral presentation Award, The Korean Physical Society
- 23/02/2016 Ulsan Mayor Award, Ulsan | Top 5 excellent graduates

Funding / Fellowships

01/01/2026 – Present	quantA postdoctoral Fellowship, qunatA
01/01/2017 – 31/12/2021	Global Ph.D. Fellowship, The Korean government Acceptance rate = 7.6:1
06/07/2014 – 18/11/2014	Global Internship Program Scholarship, The Korean Government

Supervising and mentoring activities

- Provided guidance to 4 graduates and 3 undergraduate students in research projects
- Served as teaching assistant (TA) 5 times for experimental physics laboratory courses
- Designed experiments and assisted as TA twice for physical chemistry laboratory courses
- Student representative, Department of Electrical and Electronic Engineering (2012): liaison between students and faculty, organizing departmental meetings and student activities. Later, I changed my major from Device Physics to Physics

Skills

• Laboratory Techniques

- Matter-wave scattering experiments of ^4He , D_2 at 2-100 neV under grazing incidence at high vacuum (10^{-6} to 10^{-12} mbar)
- Maintained vacuum system: Diffusion, rotary, NEG and turbo pumps
- Operation, maintenance, and troubleshooting of nanosecond infrared (Nd:YAG laser: surelite, powerlite) and visible dye (Sirah) lasers
- Molecular beams and ion detection techniques: Time-of-Flight (TOF) mass spectrometry, Velocity Map Ion Imaging (VMI), Resonance Enhanced Multiphoton Ionization (REMPI)
- Atomic Force Microscopy (AFM: Dimension ICON, NX10): Image acquisition, processing, and analysis
- Authored operating manuals and standard procedures for daily laboratory experiment

• Computational Skills

- Molecular trajectory calculations with Monte-Carlo method by solving Schrödinger equation (C^{++})
- Experimental automation using LabVIEW programming
- Converted and upgraded simulations originally created in other programming languages (C, Python, Mathematica) into languages compatible with the lab I worked in (C^{++} , Matlab, LabVIEW)
- C^{++} , Matlab, LabVIEW, LaTeX, Origin (Analyzation Software), Adobe Illustrator

• Languages

- Korean (native), English (fluent; working language of research and publication), German (intermediate, B1 TELC certificate)

Conference Activities

07/09/2026	Germany	Oral	Hot topic , 25 th European Conference on the dynamics of Molecular Systems
02/06/2026	Austria	Poster	Conference on Quantum Fluid Clusters, QFC 2026
31/05/2026	Austria	Oral	COST COSY WG4 Meeting
23/06/2025	Switzerland	Poster	2025 International Symposium on Molecular Beams (ISMB)
01/07/2024	Poland	Poster	24 th European Conference on the dynamics of Molecular Systems
22/10/2021	Korea	Poster	2021 Korean Physical Society Fall Meeting
01/07/2021	Online	Oral	Invited talk , 2021 International Symposium on Molecular Beams (ISMB)
23/04/2021	Korea	Poster	2021 Korean Physical Society Spring Meeting
10/03/2021	Online	Oral	The 5th Asian Workshop on Molecular Spectroscopy (AWMS)
26/11/2020	Korea	Oral	2020 Society of Global Ph.D. Fellows Annual Conference
06/11/2020	Korea	Oral	2020 Korean Physical Society Fall Meeting
21/11/2019	Korea	Oral	2019 Society of Global Ph.D. Fellows Annual Conference
24/10/2019	Korea	Oral, Poster	2019 Korean Physical Society Fall Meeting
18/09/2018	Greece	Poster	2018 Frontiers of Matter Wave Optics (FOMO)
19/09/2017	Korea	Poster	2017 Korean-French Joint Symposium
19/04/2017	Korea	Oral	2017 Korean Physical Society Spring Meeting
19/10/2016	Korea	Poster	2016 Korean Physical Society Fall Meeting
23/08/2016	Norway	Poster	Scattering of Atoms and Molecules from Surfaces (SAMS)
21/04/2016	Korea	Oral	2016 Korean Physical Society Spring Meeting
18/02/2016	Korea	Oral	The 13 th Symposium on Gas-phase chemical dynamics & spectroscopy
17/08/2015	Korea	Oral	The 12 th Symposium on Gas-phase chemical dynamics & spectroscopy
30/06/2015	Spain	Poster	2015 Symposium on Molecular Beams (ISMB)
23/04/2015	Korea	Oral	2015 Korean Physical Society Spring Meeting
16/04/2015	Korea	Oral	2015 Korean Chemical Society Spring Meeting
26/06/2014	Korea	Poster	The 116 th 2014 Korean Physical Chemistry Society Fall Meeting

List of Publications

1. **Lee Yeong Kim**[†], Do Won Kang[†], Jong Chan Lee, Eunmi Chae, Wieland Schöllkopf, and Bum Suk Zhao “Comprehensive peak-width analysis in matter-wave diffraction under grazing incidence conditions” *Phys. Rev. A* **110**, 013313 (2024). DOI: <https://doi.org/10.1103/PhysRevA.110.013313>
2. **Lee Yeong Kim**[†], Do Won Kang[†], Sanghwan Park, Seongyeop Lim, Jangwoo Kim, Wieland Schöllkopf, and Bum Suk Zhao “Diffractive mirrors for neutral-atom matter-wave optics” *Farad. Disc.* **251**, 160 (2024). DOI: <https://doi.org/10.1039/D3FD00155E>
3. **Lee Yeong Kim**, Sanghwan Park, Chang Young Lee, Wieland Schöllkopf, and Bum Suk Zhao “Enhanced elastic scattering of He₂ and He₃ from solids by multiple-edge diffraction” *Phys. Chem. Chem. Phys.* **24**, 21593 (2022). (Front cover) DOI: <https://doi.org/10.1039/D2CP02641D>
4. **Lee Yeong Kim**[†], Ju Hyeon Lee[†], Yun-Tae Kim, Sanghwan Park, Chang Young Lee, Wieland Schöllkopf, and Bum Suk Zhao “Experimental test of Babinet's Principle in matter-wave diffraction” *Phys. Chem. Chem. Phys.* **23**, 8030 (2021). DOI: <https://doi.org/10.1039/D0CP05694D>
5. **Lee Yeong Kim**[†], Byung Gwun Jin[†], Tae Woo Kim, Ju Hyeon Lee, and Bum Suk Zhao* “Scattering of adiabatically aligned molecules by nonresonant optical standing waves” *Sci. Adv.* **6**, eaaz0682 (2020). DOI: <https://doi.org/10.1126/sciadv.aaz0682>
6. Ju Hyeon Lee[†], **Lee Yeong Kim**[†], Yun-Tae Kim, Chang Young Lee, Wieland Schöllkopf and Bum Suk Zhao “Matter-wave diffraction from a periodic array of half planes” *Phys. Rev. Lett.* **122**, 040401 (2019). (Editor's Suggestion) DOI: <https://doi.org/10.1103/PhysRevLett.122.040401>
7. Weiqing Zhang[†], Ju Hyeon Lee[†], Hye Ah Kim, Byung Gwun Jin, Bong Jun Kim, **Lee Yeong Kim**, Bum Suk Zhao, and Wieland Schöllkopf “Universal diffraction of atomic and molecular matterwaves: A Comparison of He and D₂ quantum reflected from a grating” *Chem. Phys. Chem.* **17**, 3670 (2016). DOI: <https://doi.org/10.1002/cphc.201600965>
8. Xing Nan Sun[†], Byung Gwun Jin[†], **Lee Yeong Kim**, Bong Jun Kim, and Bum Suk Zhao “Strong optical dipole force exerted on molecules having low rotational temperature” *Chem. Phys. Chem.* **17**, 3701 (2016). DOI: <https://doi.org/10.1002/cphc.201600838>
9. **Lee Yeong Kim**, Ju Hyeon Lee, Hye Ah Kim, Sang Kyu Kwak, Bretislav Friedrich, and Bum Suk Zhao “Effect of rotational-state-dependent molecular alignment on the optical dipole force” *Phys. Rev. A* **94**, 013428 (2016). DOI: <https://doi.org/10.1103/PhysRevA.94.013428>
10. Xing Nan Sun[†], **Lee Yeong Kim**[†], Bum Suk Zhao, and Doo Soo Chung “Rotational-state-dependent dispersion of molecules by pulsed optical standing waves” *Phys. Rev. Lett.* **115**, 223001 (2015). DOI: <https://doi.org/10.1103/PhysRevLett.115.223001>

[†]Authors contributed equally to this work